

Flare Timing Reveals a Hidden Black Hole Clock

Recurrence time scales linearly with flare rise time across QPEs and repeated nuclear transients.

arXiv:2609.08737 — QPE and recurrent nuclear transient timescale scaling relations

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Background

Quasi-periodic eruptions and recurrent nuclear transients are repeating, high-amplitude flares from galactic nuclei. Their characteristic timescales have remained poorly understood.

No empirical framework linked flare recurrence to black hole mass or to the flare's own rise time.

What this paper is selling

Compiling QPEs and recurrent nuclear transients, the authors find recurrence time scales almost linearly with flare rise time, spanning hours to years. This offers a practical, unified way to characterize their temporal behavior.

Key results to remember

- Recurrence time versus black hole mass is positive but highly scattered.
- Recurrence time scales nearly linearly with flare rise time.
- One relation spans hours to years across both flare classes.

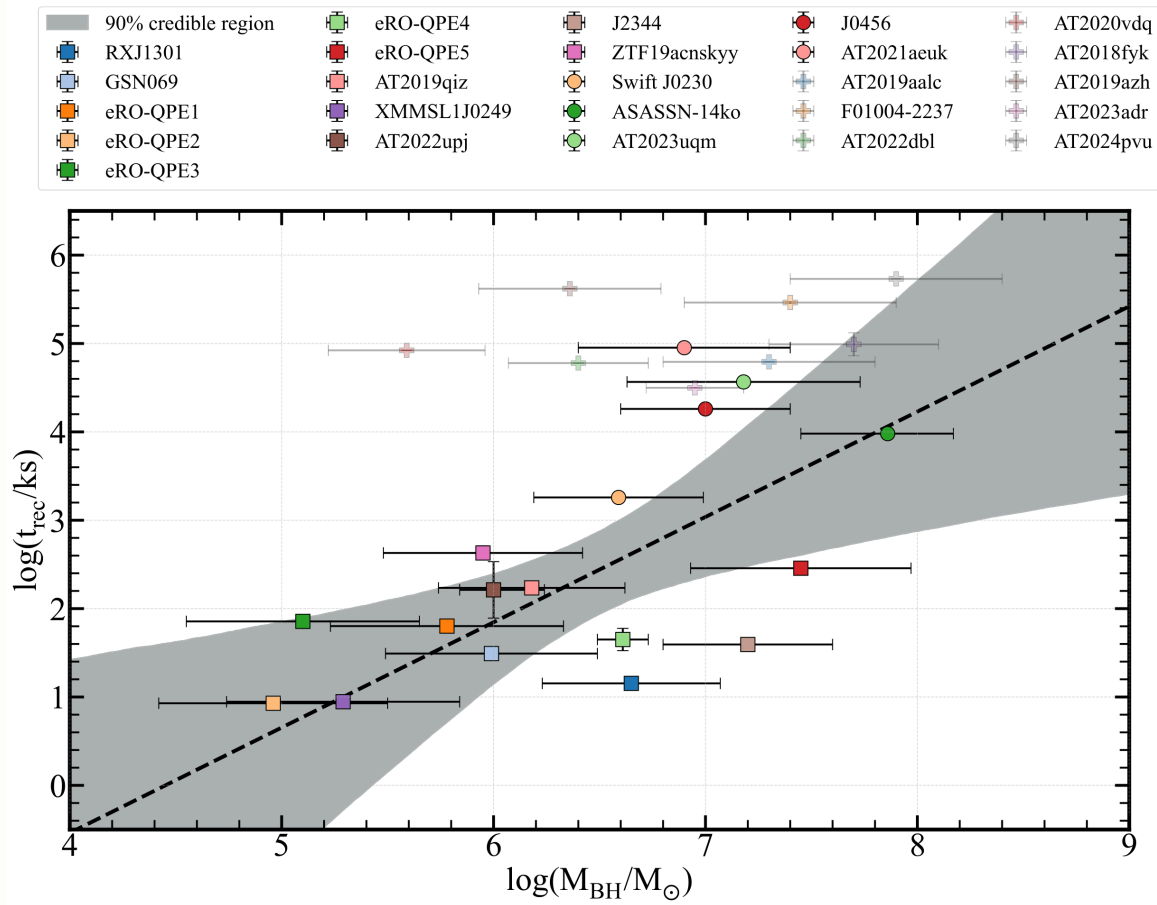


Figure 1 | Recurrence time rises with black hole mass, but with large scatter across QPEs and recurrent nuclear transients.

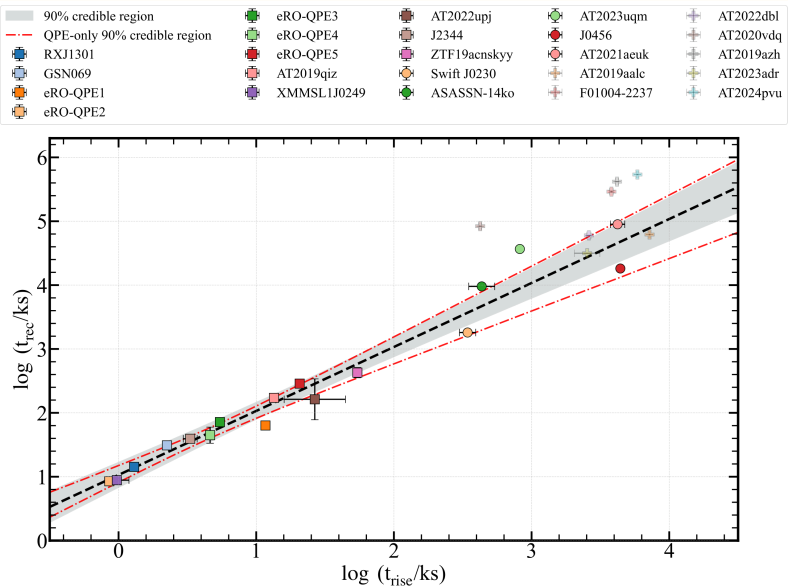


Figure 2 | A tight nearly linear rise-time versus recurrence-time relation unifies QPEs and recurrent nuclear transients.